

# Waiz Khan

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## EDUCATION

### Johns Hopkins University

*M.S.E. in Data Science*

Baltimore, MD

Aug. 2024 – May 2026

- Coursework: Investment Science, Financial Derivatives, Deep Learning, Artificial Intelligence, Statistical Learning

### GD Goenka University

*B.Tech. in Computer Science & Engineering*

Gurugram, India

Aug. 2019 – Nov. 2023

- Coursework: Operations Research, Numerical Analysis, Stochastic Processes, Linear Algebra, Data Mining, Distributed Systems

## EXPERIENCE

### Research Assistant

*Johns Hopkins University, Center for Language and Speech Processing*

Dec. 2025 – Present

Baltimore, MD

*Advisor: Dr. Philipp Koehn*

- Developing a multimodal neural machine translation system for ancient Egyptian hieroglyphs, engineering hybrid encoder architectures to fuse visual representations with symbolic sequences in extreme low-resource settings
- Designing custom preprocessing pipelines for high-noise domains; benchmarking representation learning strategies across vision and sequence models to address data scarcity and improve translation quality

### Research Assistant

*GD Goenka University*

Aug. 2022 – Nov. 2022

Gurugram, India

*Advisor: Dr. Yogesh Kumar*

- Built end-to-end ML, causal inference, and optimization pipeline for educational risk prediction on 47,027 records, achieving 84.1% ROC-AUC with DoWhy causal validation
- Developed ILP allocator that doubled expected impact per budget dollar; stress-tested with 10,000 Monte Carlo simulations and VaR reporting

### Data Scientist Intern

*EBMS Solutions Pvt Ltd*

May 2022 – Sep. 2022

Chennai, India

- Engineered churn prediction pipeline using XGBoost with time-series CV; deployed to production CRM with automated retraining; achieved +15% retention lift and AUC 0.82
- Built self-serve analytics layer with curated cohort tables and data quality monitoring; reduced reporting cycle time by 30%

## PROJECTS

### Black-Scholes Options Pricer | *Python, NumPy, SciPy, Streamlit*

Jan. 2025

- Built production-grade European options pricer with Greeks calculation (Delta, Gamma, Vega, Theta, Rho) and implied volatility solver using Newton-Raphson with Brent fallback
- Implemented arbitrage diagnostics (put-call parity, butterfly spreads, strike monotonicity); 91+ tests, 85% coverage

### Monte Carlo Risk Engine | *Python, pandas, Plotly, Streamlit*

Jan. 2025

- Developed portfolio risk simulator computing VaR, CVaR, Sharpe, Sortino, and Max Drawdown using multivariate normal and GBM models
- Applied Ledoit-Wolf covariance shrinkage and walk-forward validation to prevent look-ahead bias; 90%+ test coverage

## TECHNICAL SKILLS

**Languages:** Python, R, SQL, Rust, Swift, Java, C++, JavaScript, Bash

**ML/AI:** PyTorch, TensorFlow, scikit-learn, XGBoost; CNNs, transformers, NLP; causal inference (DoWhy); time-series CV

**Quant/Finance:** Black-Scholes, Greeks, VaR/CVaR/ES; Monte Carlo simulation; portfolio optimization; backtesting

**Data/Cloud:** pandas, NumPy, SciPy; PostgreSQL, Spark; AWS (EC2, S3, SageMaker); Git, Docker; Tableau, Power BI

**Security/iOS:** Digital forensics, memory/log analysis, incident response; SwiftUI, Core ML, HealthKit